Detecting Power Plant NO_x Emissions Changes using Physics-Guided Deep Learning and TEMPO Satellite Imagery

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Abstract

NASA's geostationary Tropospheric Emissions: Monitoring of Pollution (TEMPO) instrument provides hourly observations of tropospheric nitrogen dioxide (NO_2) across North America. Reliable satellite-based estimates of power plant nitrogen oxides (NO_x) emissions from these observations could fill gaps in reported data caused by continuous emissions monitoring system (CEMS) failures and provide independent estimates for facilities not subject to hourly emissions-reporting requirements. Previous studies have estimated hourly point-source NO_x emissions using satellite-based empirical methods, such as cross-sectional flux and the directional derivative approach. However, these methods struggle to generalize across many observations due to varying atmospheric conditions, coarse spatial resolution of TEMPO observations, and confounding signals from nearby emitting sources. We formulate the problem as change detection over sequences of TEMPO observations, allowing the model to learn plume evolution while external confounders are near-constant. Our convolutional recurrent neural network (CRNN) processes five consecutive TEMPO observations, High-Resolution Rapid Refresh (HRRR) meteorological fields, and physics-guided transport residuals to jointly classify plant emissions as increasing, steady, or decreasing and predict class-conditioned spatial contribution rasters. On a held-out evaluation set covering more than 15,000 emission sequences from over 100 power plants across the United States from August 2023 to July 2026, our model improved classification accuracy by X% over a cross-sectional flux (CSF) baseline and Y% over a seasoanlity-only basline, suggesting our model posseses stronger generalization capabilities for detecting changes in power plant emissions behavior.

1 Introduction

Based on U.S. Environmental Protection Agency (EPA) inventory data, power plants in the U.S. electricity-generation sector emitted approximately 0.74 million short tons of NO_x in 2023, about 9.5% of national emissions [13]. NO_x emissions contribute to ground-level ozone and secondary particulate matter, pollutants that can inflame and damage the airways, reduce lung function, and aggravate asthma [11]. EPA regulations require most large U.S. fossil-fuel-fired generating units that produce electricity for sale to submit unit-level hourly emissions data on a quarterly basis [14]. However, monitoring equipment malfunctions and maintenance can make CEMS measurements unavailable or invalid, requiring plants to report calculated substitute values instead of direct measurements [12]. Behind-the-meter generators that do not sell electricity can fall outside federal hourly reporting requirements, as illustrated by the natural-gas turbines powering SpaceXAI's Colossus 2 data center [4]. Outside the United States, comparable public hourly, unit-level emissions data are generally unavailable, even where facilities continuously monitor emissions [5]. Independent estimates could therefore fill reporting gaps and reveal short-term changes before the EPA's quarterly release of hourly emissions data.

TEMPO is an ultraviolet-visible spectrometer hosted aboard a geostationary satellite that observes tropospheric vertical column densities (TVCDs) for various trace gases across North America during daylight hours [16]. Earlier polar-orbiting instruments, including the Ozone Monitoring Instrument (OMI) and TROPOspheric Monitoring Instrument (TROPOMI), typically observe a given location once per day; TEMPO's geostationary view provides approximately hourly scans with finer spatial resolution [6, 15, 16]. Hourly sampling captures intraday changes in power plant plumes and atmospheric transport that daily observations can miss. However, power plant signals in TEMPO NO_2 TVCDs can be weakened or obscured by factors including evolving meteorology and atmospheric chemistry, cloud cover, viewing geometry and retrieval uncertainty, dilution within TEMPO's 2.0-by-4.75 km spatial resolution per pixel, and interference from nearby emission sources [4, 5, 10]. Cross-comparisons with CEMS data, which is the most reliable ground truth available, introduce an additional temporal mismatch because TEMPO observes a facility at a specific time during each hourly scan, whereas CEMS data aggregate emissions over an entire clock hour [10, 14].

Satellite-based point-source studies typically estimate emissions by relating an observed column enhancement to atmospheric transport. The cross-sectional flux (CSF) method integrates background-subtracted columns across plume transects and multiplies the resulting line density by an effective wind speed, whereas the integrated mass enhancement (IME) method converts the total excess plume mass to a source rate using an estimated residence time [7, 8]. Gaussian plume methods instead fit a parameterized dispersion model to the observed enhancement [8]. Flux divergence and directional derivative approaches (DDA) apply mass-conservation relationships to satellite column fields and meteorological data without first fitting a plume shape. These approaches have been applied to TROPOMI NO_2 observations to estimate point-source NO_x emissions [7, 9]. Sun et al. extended DDA with a column-tendency term and used TEMPO observations to derive hourly estimates for 14 U.S. power plants [9]. Machine-learning studies of satellite NO_2 have used convolutional neural networks to identify TROPOMI plumes and classifiers to predict whether power-plant plumes are detectable under given meteorological, surface, and viewing conditions [3, 5]. These studies identify when a plume is visible but do not infer hourly plant emissions from sequences of NO_2 imagery. For other trace gases, learned methods have progressed from plume segmentation to source-rate estimation, including a U-Net

and CNN pipeline for methane and CNN models trained to estimate power-plant CO_2 emissions from simulated satellite scenes [1, 2].

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